WoSM Academy

MAT231-F27 · Calculus II

The Gaussian integral

and improper multiple integrals

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Lecture 21 · Unit 21 · Fall 2027

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Last time

- Change of variables: $\iint_D f\,dA = \iint_S f(T(u,v))\,\left|\frac{\partial(x,y)}{\partial(u,v)}\right|\,du\,dv$.
- Polar $dA = r\,dr\,d\theta$; cylindrical $dV = r\,dr\,d\theta\,dz$; spherical $dV = \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta$.

Today polar coordinates compute an integral in *one* variable that no antiderivative can: the area under the bell curve. It needs integrals over the whole plane, so we also make sense of improper multiple integrals.

This lecture

01 The Gaussian integral

02 Improper multiple integrals

03 Probability and geometry

04 Summary and practice

Reading: Marsden & Tromba, *Vector Calculus*, Section 6.4.

01

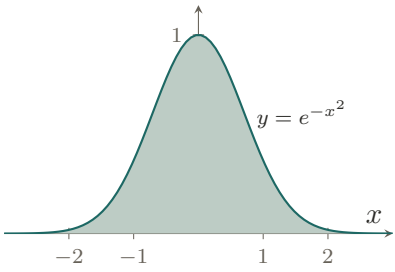
The Gaussian integral

The bell curve

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$$

- It converges: for $|x| \geq 1$, $e^{-x^2} \leq e^{-|x|}$ (comparison, Calculus I, Unit 23).
- e^{-x^2} has no elementary antiderivative (a theorem of Liouville), so the Fundamental Theorem of Calculus does not help.
- Numerically $I \approx 1.7725$.

The value is $\sqrt{\pi} \approx 1.77245$, and the proof goes through **two** dimensions.



The shaded area is $\sqrt{\pi}$.

The trick: square it

The name of the variable does not matter, so

$$I^2 = \Big(\int_{-\infty}^{\infty} e^{-x^2}\,dx\Big)\Big(\int_{-\infty}^{\infty} e^{-y^2}\,dy\Big) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dA.$$

The integrand depends only on r. In polar coordinates:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta = 2\pi\Big[-\frac{1}{2}e^{-r^2}\Big]_0^{\infty} = \pi.$$

The factor r from dA is exactly what makes e^{-r^2} integrable in closed form. So

$$\boxed{\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}}$$

Two steps need justification: integrals over all of $\mathbb{R}^2$, and the polar change of variables for them. Next.

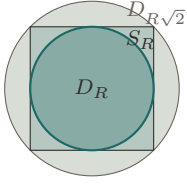
Making it rigorous

Let D_R be the disc of radius R and $S_R = [-R, R]^2$. Then $D_R \subset S_R \subset D_{R\sqrt{2}}$, and the integrand is positive, so

$$\iint_{D_R} e^{-(x^2+y^2)}\,dA \;\le\; \Big(\int_{-R}^{R} e^{-x^2}\,dx\Big)^2 \;\le\; \iint_{D_{R\sqrt{2}}} e^{-(x^2+y^2)}\,dA.$$

The middle term is the double integral over S_R (Fubini). The outer ones are proper integrals, computed in polar coordinates:

$$\iint_{D_R} e^{-(x^2+y^2)}\,dA = \pi\big(1 - e^{-R^2}\big) \;\to\; \pi.$$



By the squeeze theorem, $\big(\int_{-R}^{R} e^{-x^2}\,dx\big)^2 \to \pi$. □

Consequences

- Substituting $x = \sqrt{a}\,u$: $\displaystyle\int_{-\infty}^{\infty} e^{-ax^2}\,dx = \sqrt{\frac{\pi}{a}}$ for $a > 0$.
- With $a = \frac{1}{2}$: $\displaystyle\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}$. This is the $\sqrt{2\pi}$ in Stirling's formula $n! \sim \sqrt{2\pi n}\,(n/e)^n$ (Lecture 8).
- The **standard normal density** $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ has total integral 1; that is where its constant comes from.
- Moments, by parts: $\displaystyle\int_{-\infty}^{\infty} x^2 e^{-x^2/2}\,dx = \Big[-xe^{-x^2/2}\Big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}$. So the standard normal distribution has variance $\int x^2\varphi(x)\,dx = 1$.

The Gamma function

Definition

For $s > 0$: $\Gamma(s) = \displaystyle\int_0^\infty t^{s-1} e^{-t}\, dt.$

- By parts: $\Gamma(s+1) = s\,\Gamma(s)$, and $\Gamma(1) = 1$. So $\Gamma(n+1) = n!$: Γ extends the factorial to non-integers.
- With $t = x^2$, $dt = 2x\, dx$:

$$\Gamma\left(\tfrac{1}{2}\right) = \int_0^\infty x^{-1} e^{-x^2}\, 2x\, dx = 2\int_0^\infty e^{-x^2}\, dx = \sqrt{\pi}.$$

- Hence $\Gamma\left(\tfrac{3}{2}\right) = \tfrac{1}{2}\sqrt{\pi}$, $\Gamma\left(\tfrac{5}{2}\right) = \tfrac{3}{4}\sqrt{\pi}$, and so on.

Stirling's formula holds for Γ as well: $\Gamma(s+1) \sim \sqrt{2\pi s}\,(s/e)^s$ as $s \to \infty$.

02

Improper multiple integrals

Improper integrals in several variables

Definition

Let $f \geq 0$ be continuous on D. An **exhaustion** of D is a sequence of bounded regions $D_1 \subseteq D_2 \subseteq \dots$ with $\bigcup_k D_k = D$, on each of which f is integrable. Then

$$\iint_D f\,dA = \lim_{k\to\infty} \iint_{D_k} f\,dA \qquad \in [0,\infty].$$

The integral **converges** if this limit is finite.

- For $f \geq 0$ the sequence is increasing, so the limit exists, and it is the **same** for every exhaustion.
- D_k can avoid a point where f blows up, e.g. $D_k = \{\frac{1}{k} \leq \|\mathbf{x}\| \leq 1\}$. The same works in $\mathbb{R}^n$.

Functions of both signs

If f takes both signs, we require **absolute convergence**: $\iint_D |f|\,dA < \infty$. Then $\iint_D f = \iint_D f^+ - \iint_D f^-$, with $f^\pm = \max(\pm f, 0)$, and every exhaustion gives this value.

Example ($f = \sin(x^2+y^2)$ on $\mathbb{R}^2$)

Over the disc of radius R: $\displaystyle\int_0^{2\pi}\int_0^{R} \sin(r^2)\, r\, dr\, d\theta = \pi\big(1-\cos R^2\big)$.

Along $R^2 = 2k\pi$ this is 0; along $R^2 = (2k+1)\pi$ it is 2π. No limit.

As with series (Lecture 3): absolutely convergent sums can be taken in any order; conditionally convergent ones depend on the order — here, on the shape of the regions.

Power laws: when do they converge?

In polar coordinates the radial integral carries the factor r:

$$\iint_{r\ge 1} \frac{dA}{r^a} = 2\pi \int_1^\infty r^{1-a}\,dr, \qquad \iint_{r\le 1} \frac{dA}{r^a} = 2\pi \int_0^1 r^{1-a}\,dr.$$

	$\mathbb{R}$	$\mathbb{R}^2$	$\mathbb{R}^n$
$\int_{\|\mathbf{x}\|\ge 1} \|\mathbf{x}\|^{-a}$ converges iff	$a > 1$	$a > 2$	$a > n$
$\int_{\|\mathbf{x}\|\le 1} \|\mathbf{x}\|^{-a}$ converges iff	$a < 1$	$a < 2$	$a < n$

In $\mathbb{R}^n$ the volume of the shell $r \le \|\mathbf{x}\| \le r + dr$ is proportional to $r^{n-1}\,dr$.

- $\iint_{\mathbb{R}^2} \frac{dA}{(1+x^2+y^2)^2} = 2\pi \int_0^\infty \frac{r\,dr}{(1+r^2)^2} = \pi$.
- $\iint_{r\le 1} \frac{dA}{\sqrt{x^2+y^2}} = 2\pi$, although the integrand is unbounded; $\iint_{r\le 1} \frac{dA}{x^2+y^2} = \infty$.

Fubini for improper integrals

Theorem (Tonelli and Fubini)

If $f \geq 0$, or if $\iint_{\mathbb{R}^2} |f|\, dA < \infty$, then

$$\iint_{\mathbb{R}^2} f\, dA = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\, dy\, dx = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\, dx\, dy,$$

and the same on any region.

- This justifies the first step of the Gaussian trick: $e^{-x^2-y^2} > 0$.
- Lecture 18's counterexample $\frac{x^2-y^2}{(x^2+y^2)^2}$ on $[0,1]^2$: in polar coordinates $|f| = \frac{|\cos 2\theta|}{r^2}$, and $\iint |f|\, dA = \iint \frac{|\cos 2\theta|}{r}\, dr\, d\theta = \infty$. Not absolutely convergent, so the orders may (and do) disagree.

Application: swapping the order

Compute $\displaystyle\int_0^\infty \frac{e^{-x} - e^{-2x}}{x}\,dx$.

Example (Write the integrand as an integral)

$\displaystyle\frac{e^{-x} - e^{-2x}}{x} = \int_1^2 e^{-xy}\,dy$. The function e^{-xy} is positive, so by Tonelli

$$\int_0^\infty \int_1^2 e^{-xy}\,dy\,dx = \int_1^2 \int_0^\infty e^{-xy}\,dx\,dy = \int_1^2 \frac{dy}{y} = \ln 2.$$

The same idea, writing a number as an integral and changing the order, evaluates many integrals that have no elementary antiderivative.

03

Probability and geometry

Two independent normal variables

If X, Y are independent standard normal, their joint density is

$$p(x,y) = \varphi(x)\,\varphi(y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}.$$

It depends only on r: the distribution is **rotation invariant**.

- $P(X^2+Y^2 \le R^2) = \displaystyle\int_0^{2\pi}\int_0^{R} \frac{1}{2\pi} e^{-r^2/2}\, r\, dr\, d\theta = 1 - e^{-R^2/2}$: 0.393 for $R=1$, 0.865 for $R=2$.
- Expected distance from the origin: $E\big[\sqrt{X^2+Y^2}\big] = \displaystyle\int_0^{\infty} r \cdot r\, e^{-r^2/2}\, dr = \sqrt{\frac{\pi}{2}} \approx 1.253.$

This rotation invariance underlies the Box–Muller method for simulating normal variables, and many proofs in Probability Theory and Statistics.

The volume of a ball in n dimensions

Let V_n be the volume of the unit ball in $\mathbb{R}^n$; the ball of radius r has volume $V_n r^n$.

- As a product: $\displaystyle\int_{\mathbb{R}^n} e^{-\|\mathbf{x}\|^2}\,dV = \Big(\int_{-\infty}^{\infty} e^{-x^2}\,dx\Big)^n = \pi^{n/2}$.
- In shells: the shell $r \le \|\mathbf{x}\| \le r + dr$ has volume $nV_n r^{n-1}\,dr$, so

$$\int_{\mathbb{R}^n} e^{-\|\mathbf{x}\|^2}\,dV = \int_0^{\infty} e^{-r^2}\, nV_n r^{n-1}\,dr = nV_n \cdot \tfrac{1}{2}\Gamma\big(\tfrac{n}{2}\big) = V_n\,\Gamma\big(\tfrac{n}{2}+1\big).$$

Result

$$V_n = \frac{\pi^{n/2}}{\Gamma\big(\frac{n}{2}+1\big)}. \qquad \text{Check: } V_2 = \pi, \quad V_3 = \frac{\pi^{3/2}}{\frac{3}{4}\sqrt{\pi}} = \frac{4}{3}\pi.$$

Balls in high dimensions

n	V_n	$V_n/2^n$
1	2	1
2	3.1416	0.785
3	4.1888	0.524
5	5.2638	0.164
10	2.5502	0.0025
20	0.0258	$2.5 \cdot 10^{-8}$

- V_n grows up to $n = 5$, then tends to 0: $\Gamma(\frac{n}{2} + 1)$ grows faster than $\pi^{n/2}$ (Stirling).
- $V_n/2^n$ is the fraction of the cube $[-1, 1]^n$ filled by the ball. In 20 dimensions, a random point of the cube is almost never in the ball.
- One face of the "curse of dimensionality": grids and rejection sampling fail in high dimensions, and intuition from $\mathbb{R}^3$ misleads.

04

Summary and practice

Summary

- $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$: square it, then use polar coordinates. Hence $\int e^{-ax^2} = \sqrt{\pi/a}$, the normal density, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.
- Improper integrals of $f \geq 0$: limits over exhaustions, independent of the exhaustion. Functions of both signs: require absolute convergence.
- $\|\mathbf{x}\|^{-a}$ is integrable near ∞ iff $a > n$, near $\mathbf{0}$ iff $a < n$.
- Tonelli and Fubini: iterated improper integrals may be swapped for $f \geq 0$ or absolutely integrable f.
- $V_n = \pi^{n/2}/\Gamma(\frac{n}{2}+1)$, which tends to 0.

Practice

1. Compute $\int_{-\infty}^{\infty} e^{-(x-1)^2/8}\,dx$.
2. Compute $\int_0^{\infty} x^2 e^{-x^2}\,dx$.
3. Compute $\iint_{\mathbb{R}^2} e^{-(x^2+4y^2)}\,dA$.
4. For which p does $\iint_{x^2+y^2\geq 1}(x^2+y^2)^{-p}\,dA$ converge? Find its value.
5. Compute $\iint_{x^2+y^2\leq 1}\ln(x^2+y^2)\,dA$.
6. Find $\Gamma(\frac{5}{2})$, and check the formula for V_n with $n=1$.
7. X, Y are independent standard normal. Find $P(|Y|\leq X)$.
8. Compute $\int_0^{\infty}\frac{e^{-x}-e^{-3x}}{x}\,dx$.

Answers

1. $u = (x-1)/(2\sqrt{2})$: $2\sqrt{2}\cdot\sqrt{\pi} = \sqrt{8\pi}$.
2. By parts, $\frac{1}{2}\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{4}$.
3. A product: $\sqrt{\pi}\cdot\sqrt{\pi/4} = \frac{\pi}{2}$ (or substitute $y = v/2$).
4. $2\pi\int_1^\infty r^{1-2p}\,dr$: converges iff $p > 1$, with value $\frac{\pi}{p-1}$.
5. $2\pi\int_0^1 2\ln r\cdot r\,dr = 4\pi\cdot\left(-\frac{1}{4}\right) = -\pi$.
6. $\Gamma(\frac{5}{2}) = \frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{3}{4}\sqrt{\pi}$. $V_1 = \sqrt{\pi}/\Gamma(\frac{3}{2}) = 2$: the interval $[-1, 1]$.
7. The region $|y| \le x$ is a wedge of angle $\frac{\pi}{2}$; by rotation invariance the probability is $\frac{\pi/2}{2\pi} = \frac{1}{4}$.
8. $\int_1^3 \frac{dy}{y} = \ln 3$.

Next time

Lecture 22 — Vector fields and line integrals (Unit 22)

- Vector fields: velocity, force, gradient fields
- Integrating along a curve: arc length, mass of a wire, work done by a force
- The fundamental theorem for line integrals, and conservative fields

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 6.1–6.3.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.